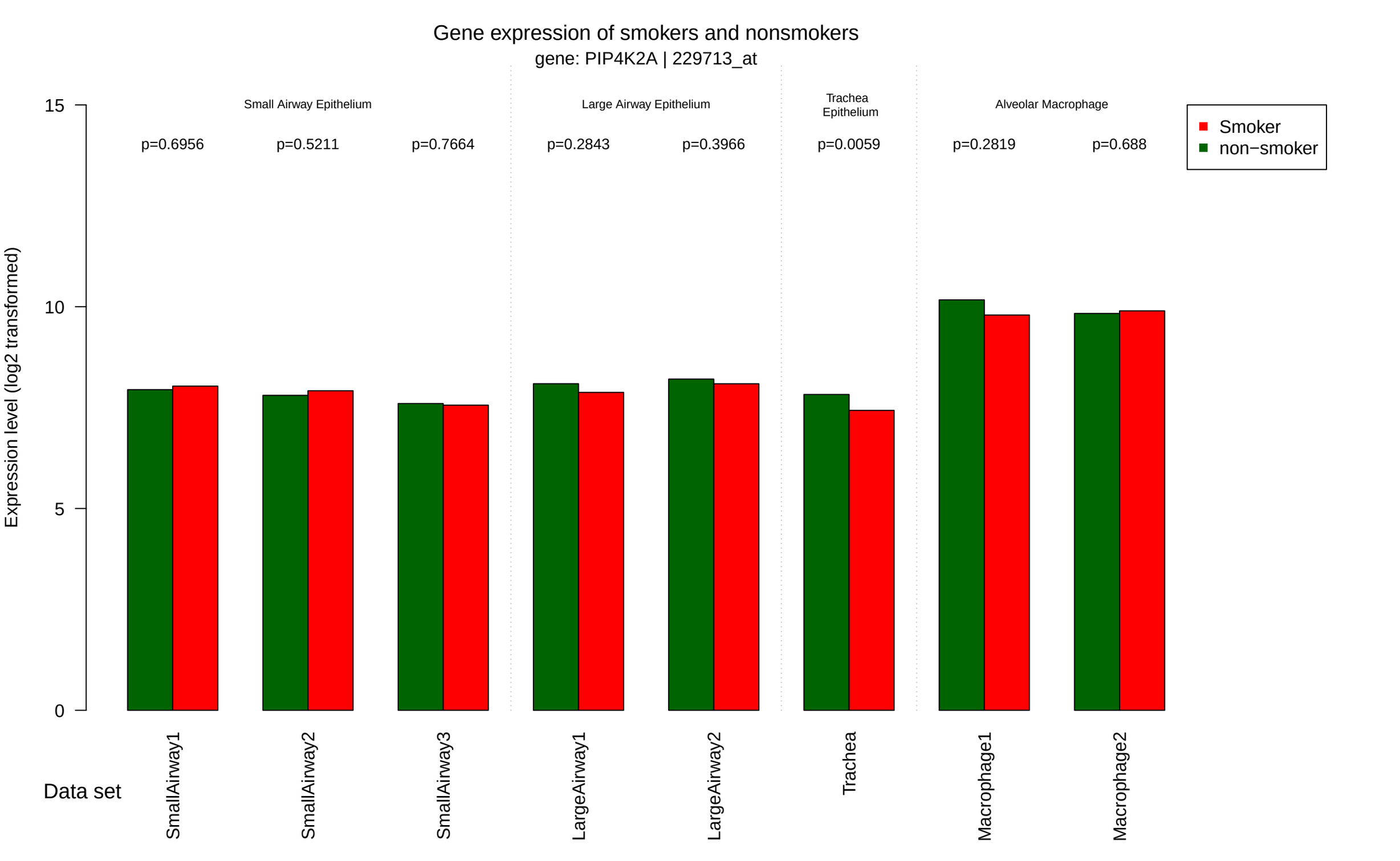
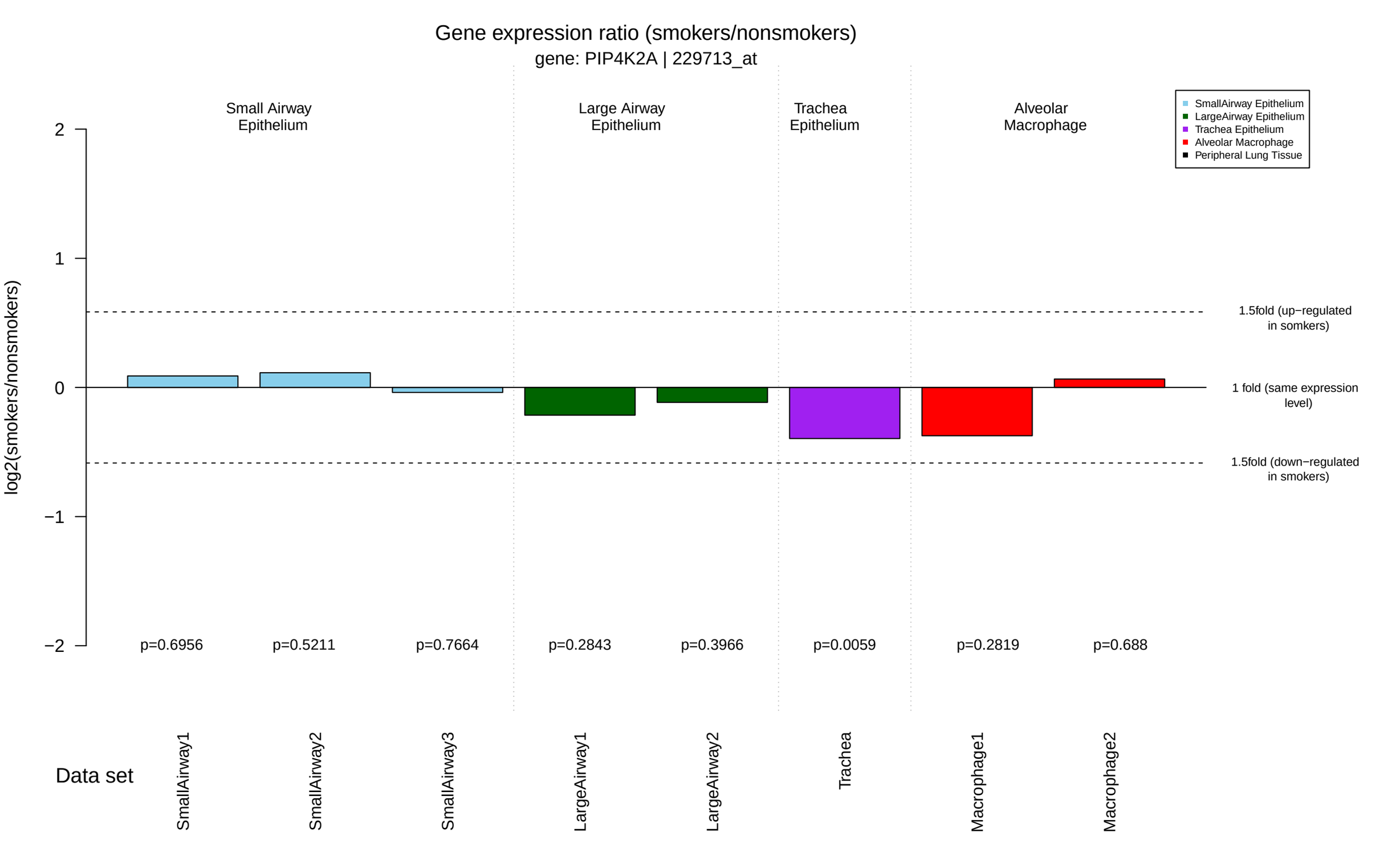
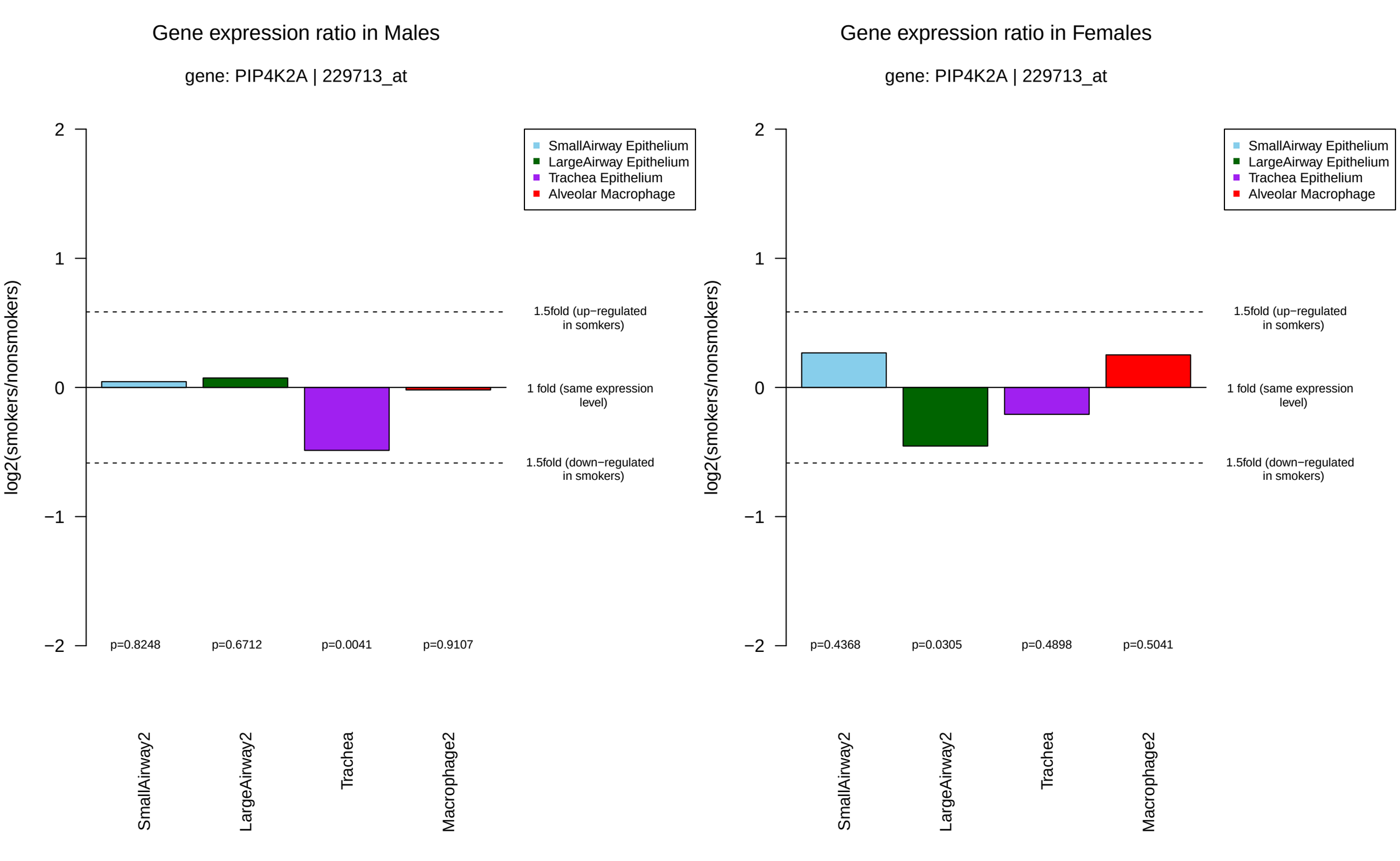
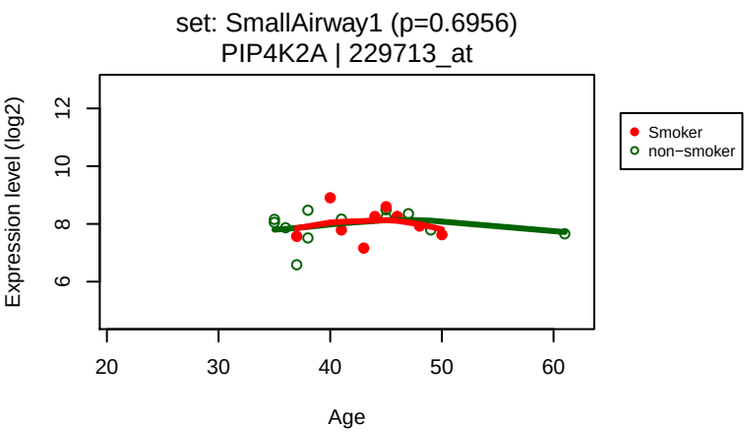
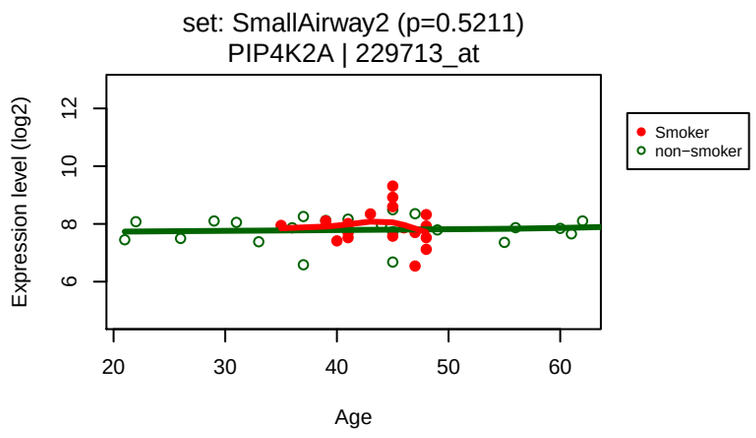
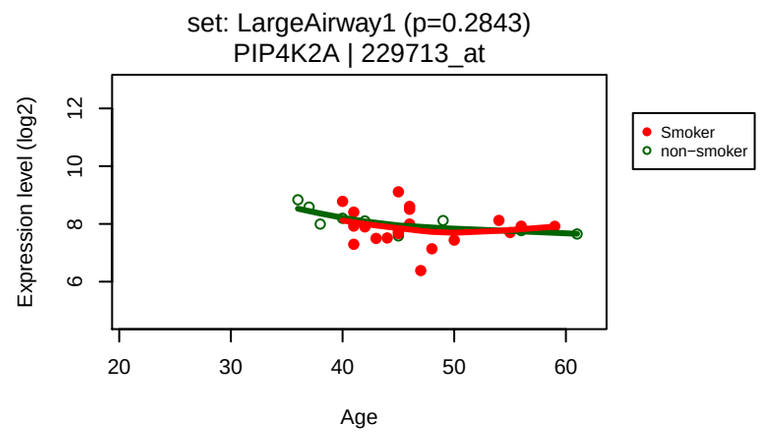

Gene expression of smokers and nonsmokers
gene: PIP4K2A | 229713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6956
p=0.5211
p=0.7664
p=0.2843
p=0.3966
p=0.0059
p=0.2819
p=0.688
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

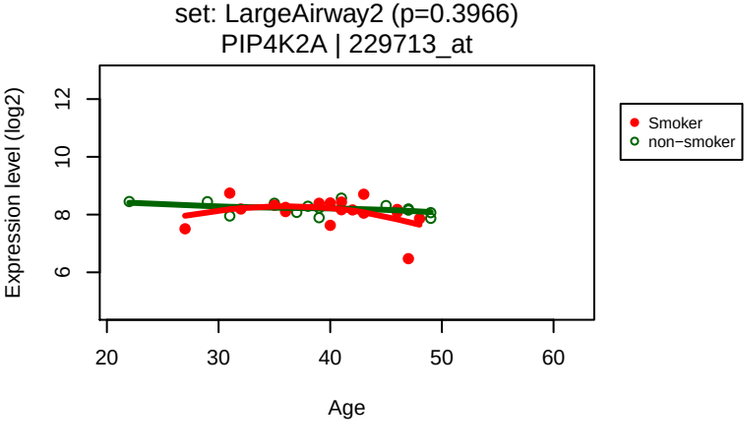

Gene expression ratio (smokers/nonsmokers)
gene: PIP4K2A | 229713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6956
p=0.5211
p=0.7664
p=0.2843
p=0.3966
p=0.0059
p=0.2819
p=0.688
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

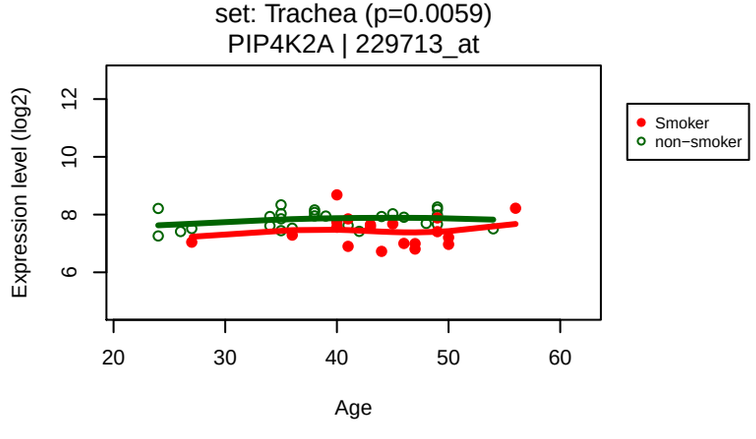

Gene expression ratio in Males
gene: PIP4K2A | 229713_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8248
p=0.6712
p=0.0041
p=0.9107
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIP4K2A | 229713_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4368
p=0.0305
p=0.4898
p=0.5041
SmallAirway2
LargeAirway2
Trachea
Macrophage2

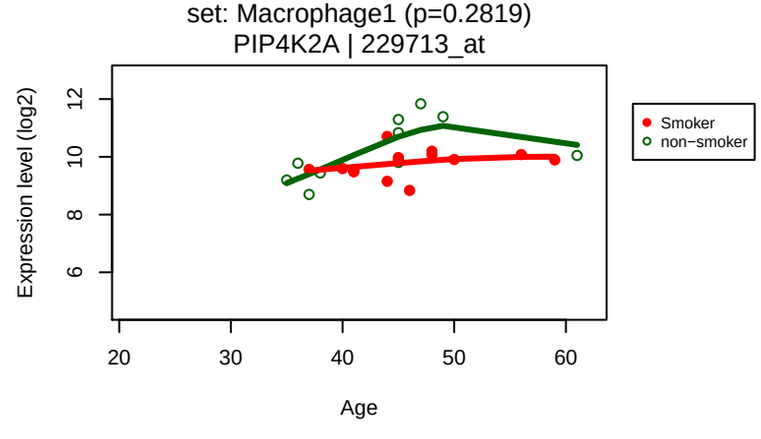

set: SmallAirway1 (p=0.6956)
PIP4K2A | 229713_at
Expression level (log2)
Age
Smoker
non−smoker

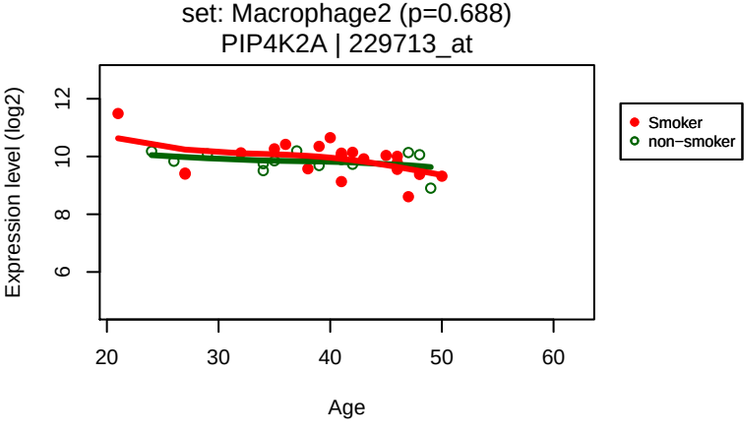

set: SmallAirway2 (p=0.5211)
PIP4K2A | 229713_at
Expression level (log2)
Age
Smoker
non−smoker

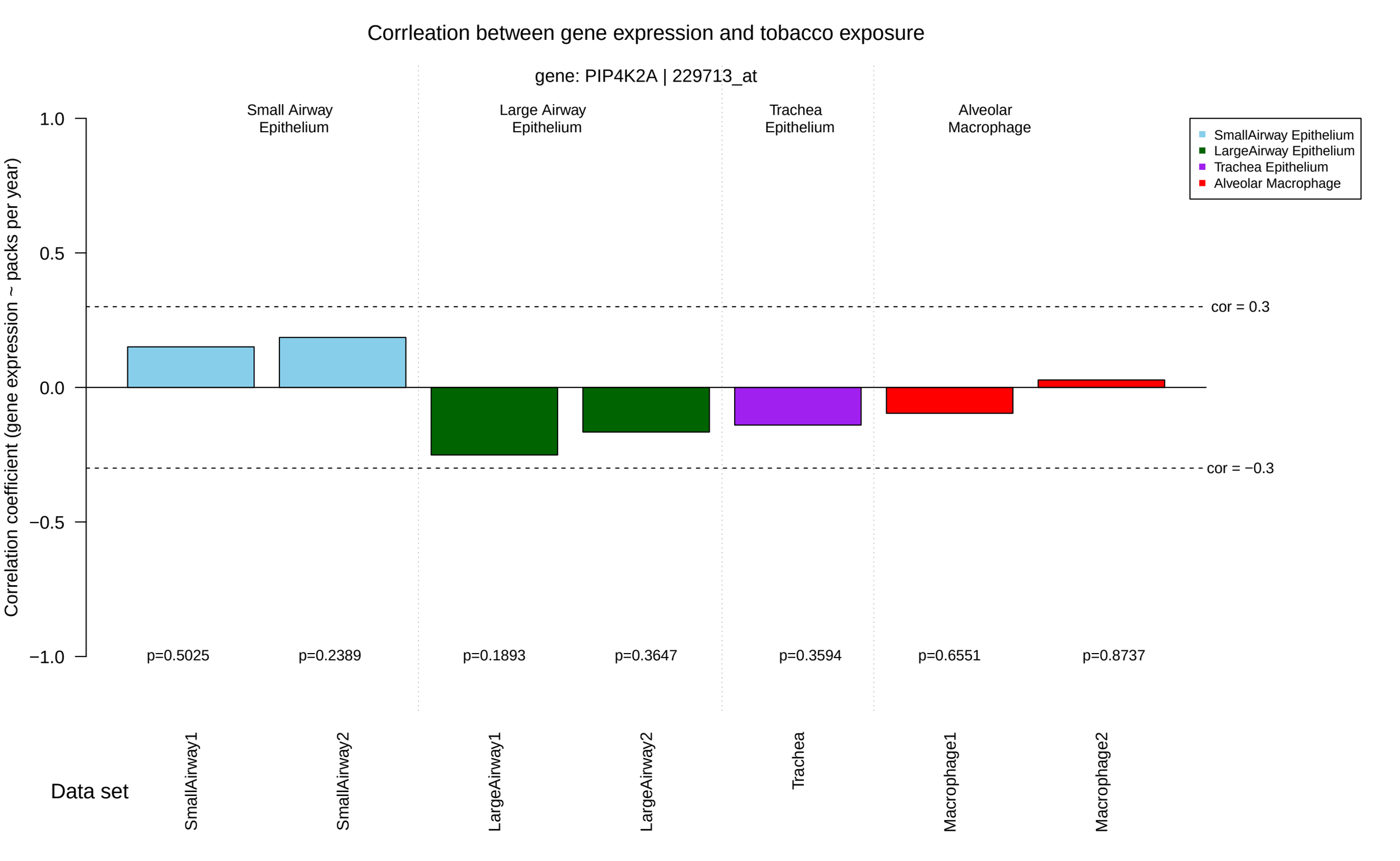

set: LargeAirway1 (p=0.2843)
PIP4K2A | 229713_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3966)
PIP4K2A | 229713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0059)
PIP4K2A | 229713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2819)
PIP4K2A | 229713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.688)
PIP4K2A | 229713_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIP4K2A | 229713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5025
p=0.2389
p=0.1893
p=0.3647
p=0.3594
p=0.6551
p=0.8737
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2